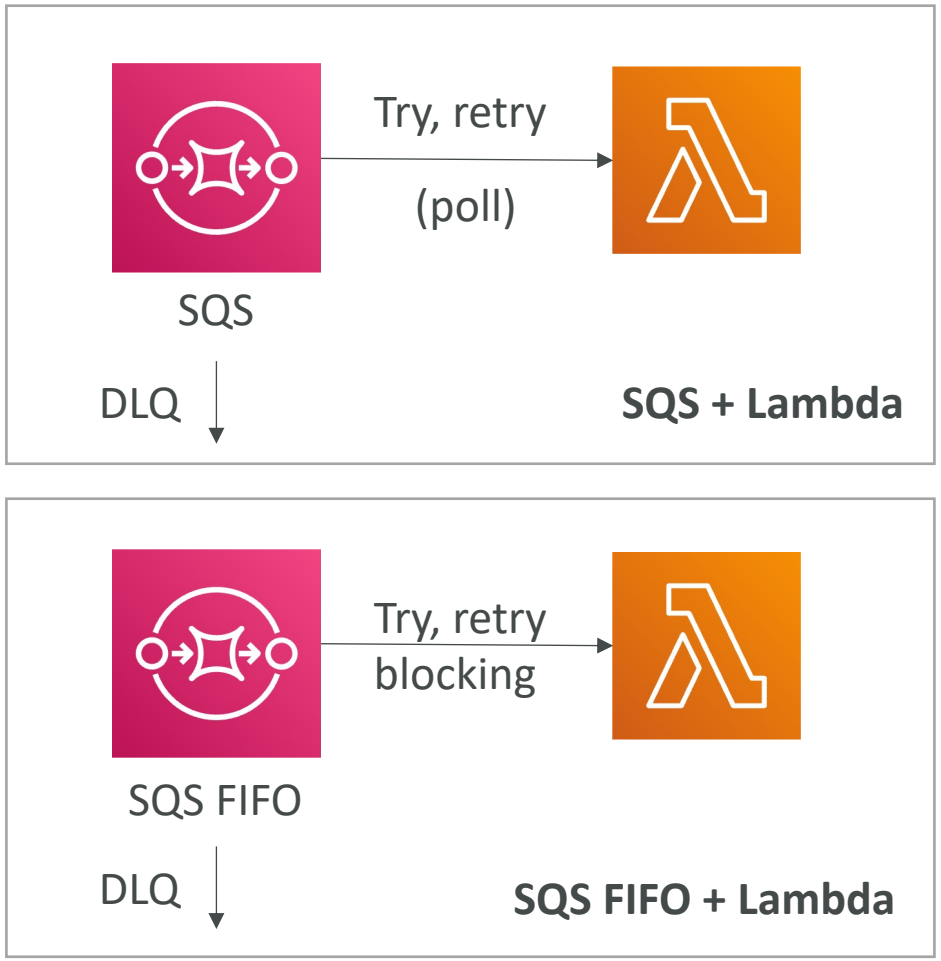
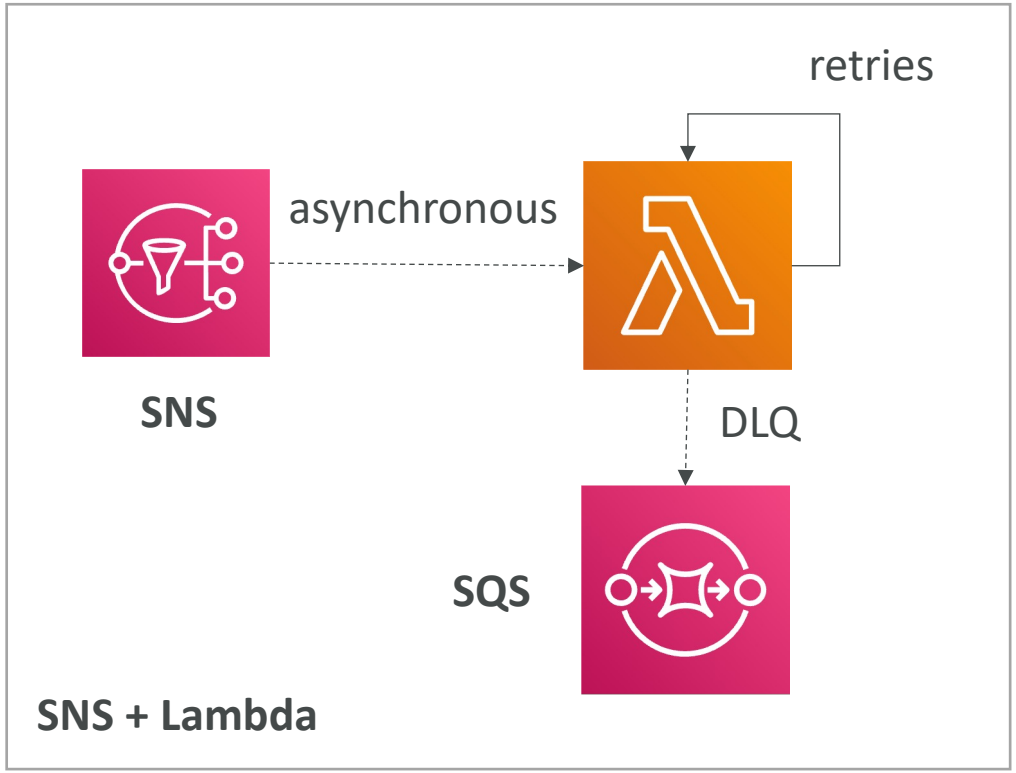
More Solutions Architecture

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Lambda, SNS & SQS

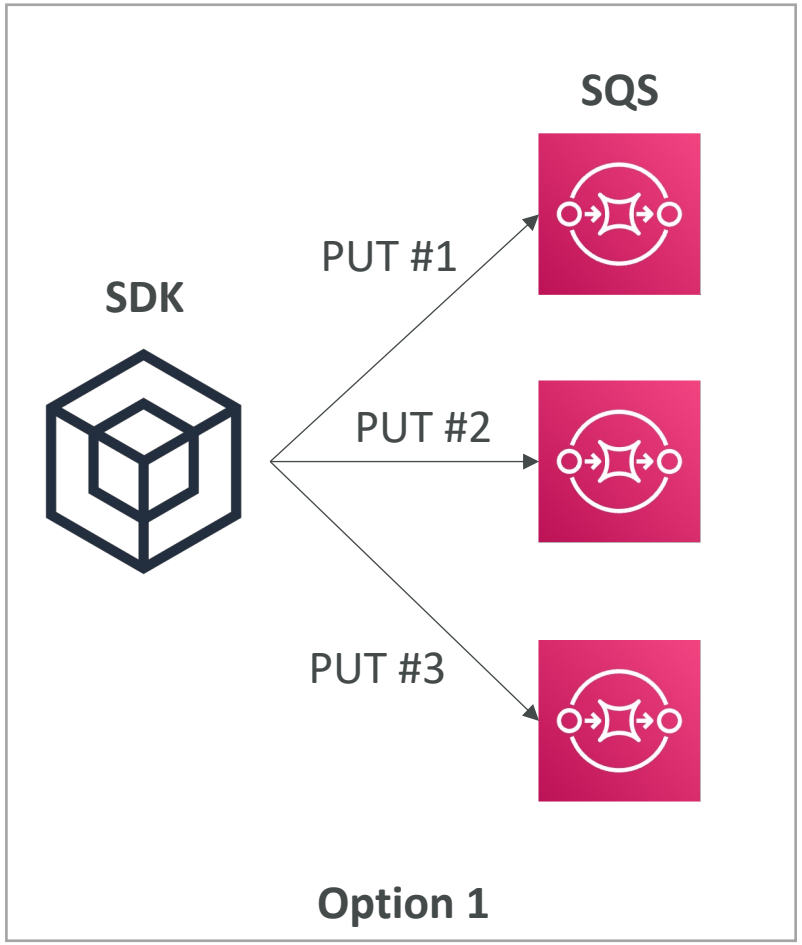


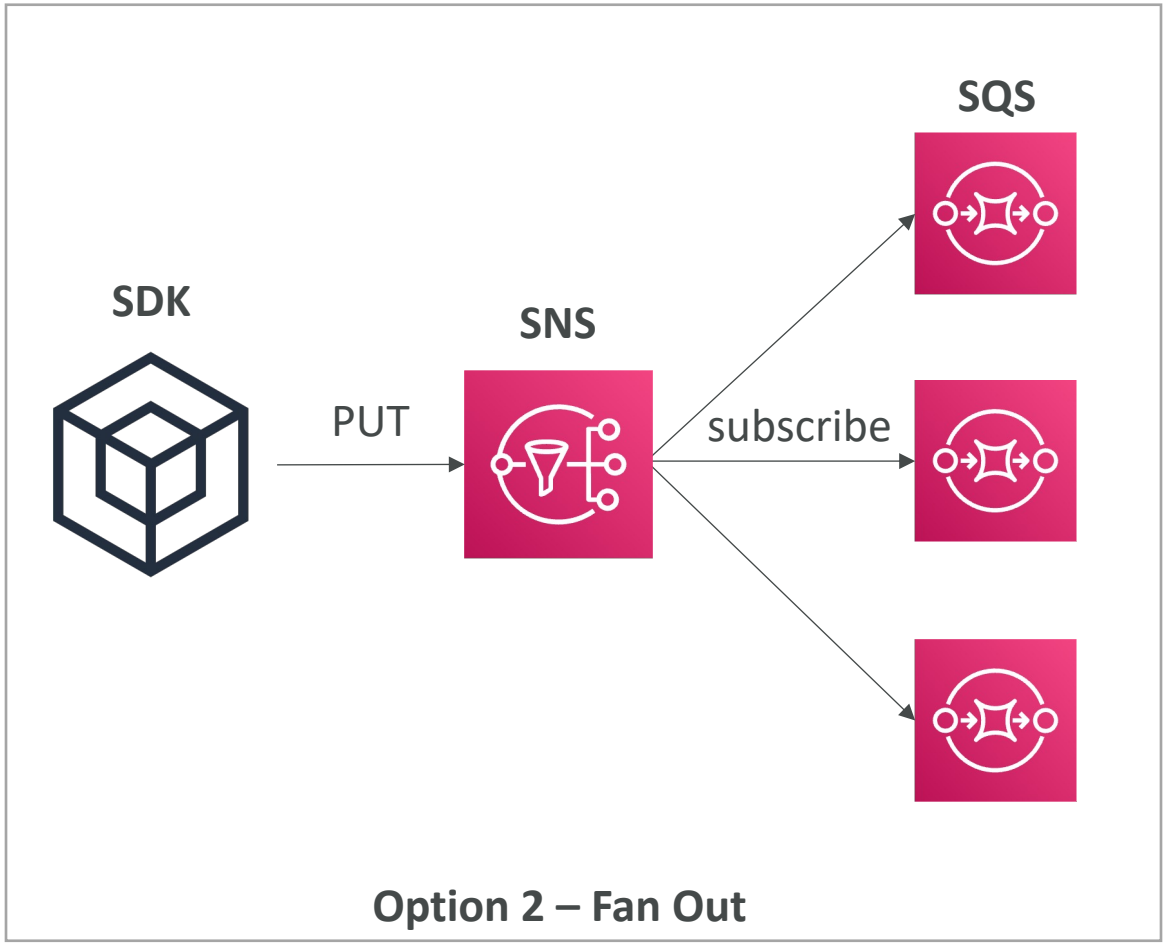


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Fan Out Pattern: deliver to multiple SQS





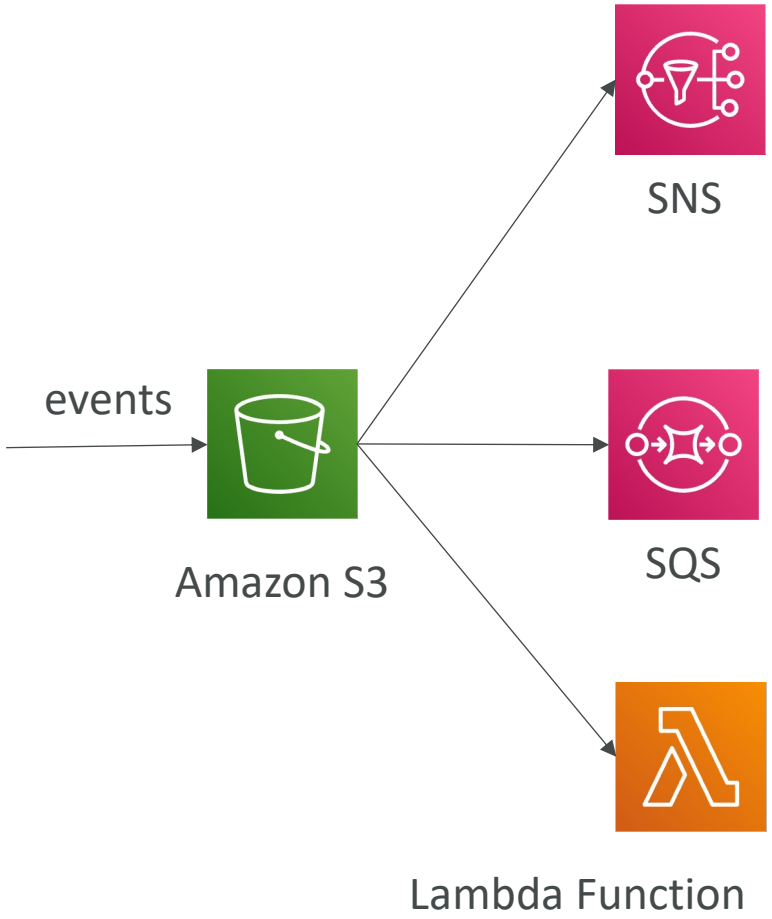
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S3 Event Notifications

- S3:ObjectCreated, S3:ObjectRemoved, S3:ObjectRestore, S3:Replication…
- Object name filtering possible (*.jpg)
- Use case: generate thumbnails of images uploaded to S3
- **Can create as many "S3 events" as desired**

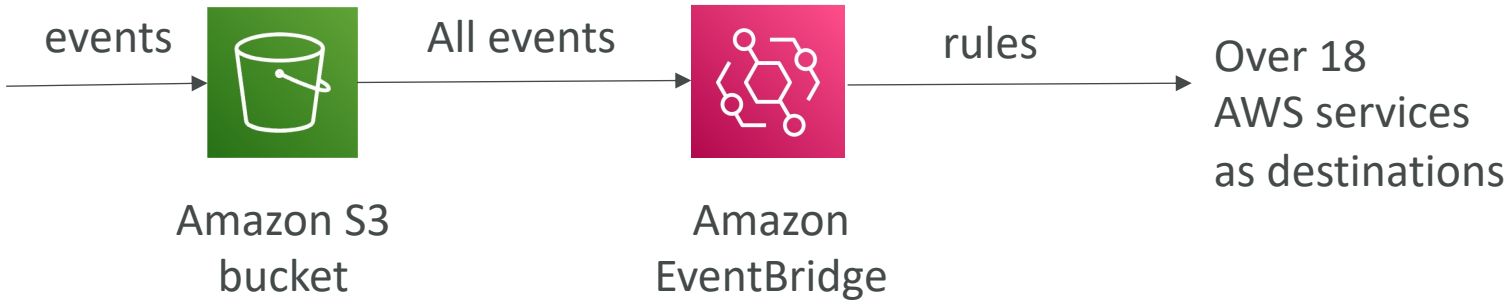
- S3 event notifications typically deliver events in seconds but can sometimes take a minute or longer



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S3 Event Notifications with Amazon EventBridge

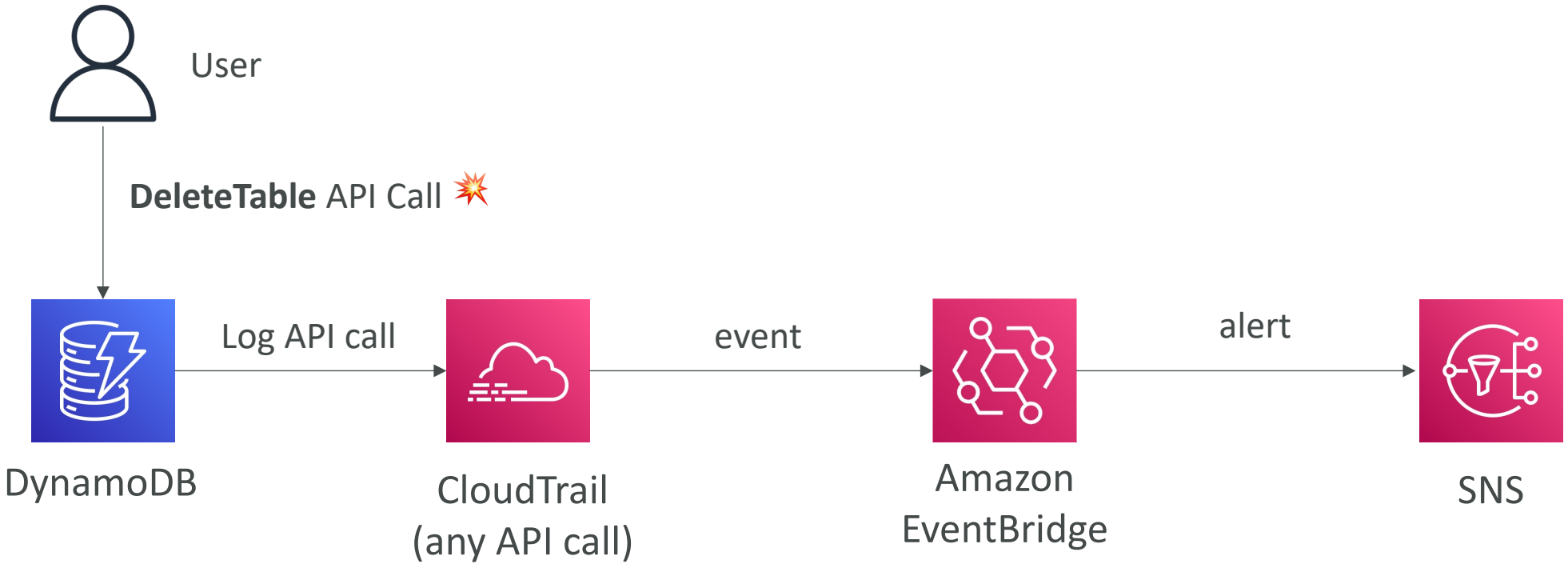


- **Advanced filtering** options with JSON rules (metadata, object size, name...)
- **Multiple Destinations** – ex Step Functions, Kinesis Streams / Firehose…
- **EventBridge Capabilities** – Archive, Replay Events, Reliable delivery

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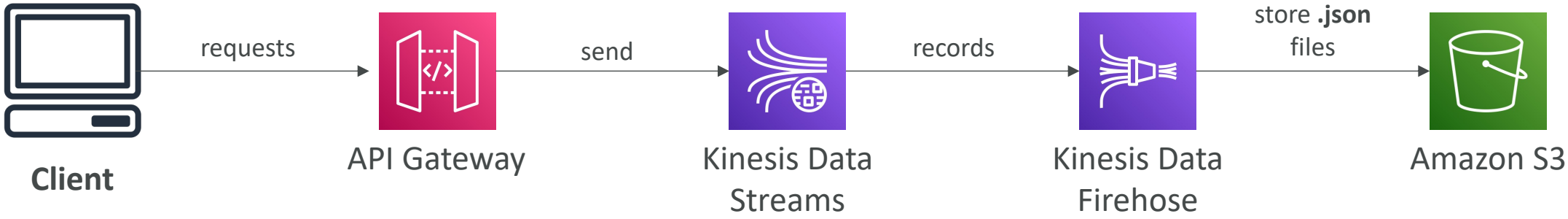
Amazon EventBridge – Intercept API Calls



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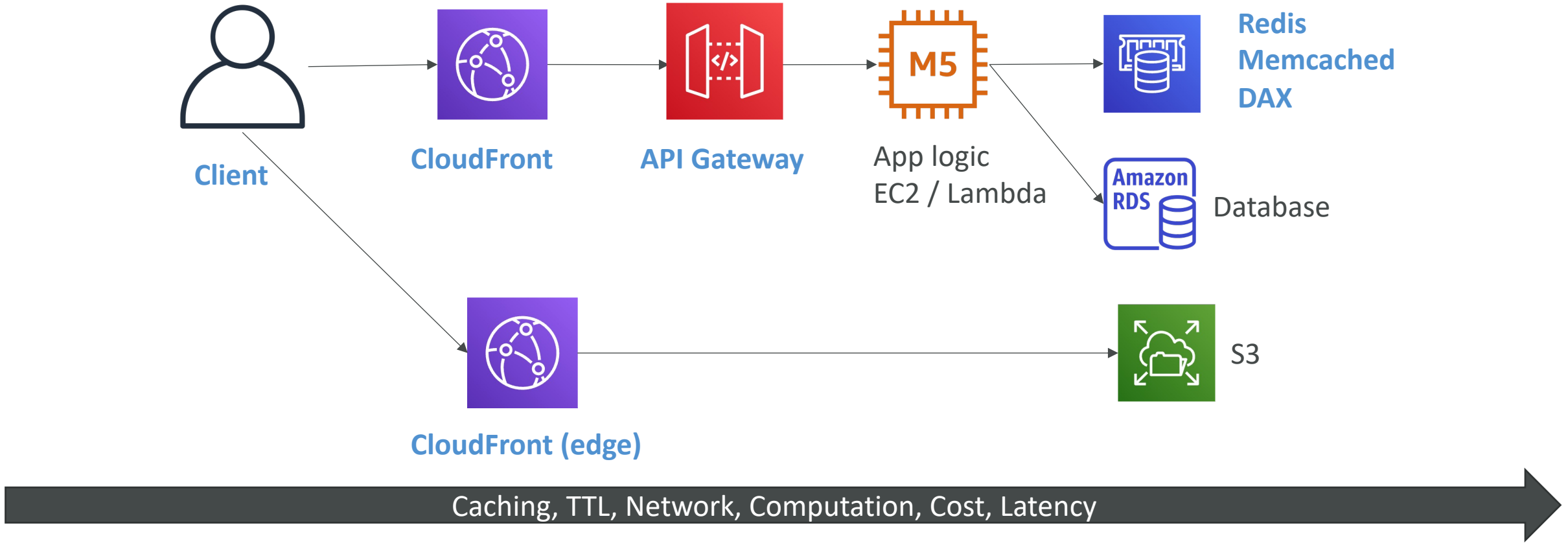
API Gateway – AWS Service Integration Kinesis Data Streams example



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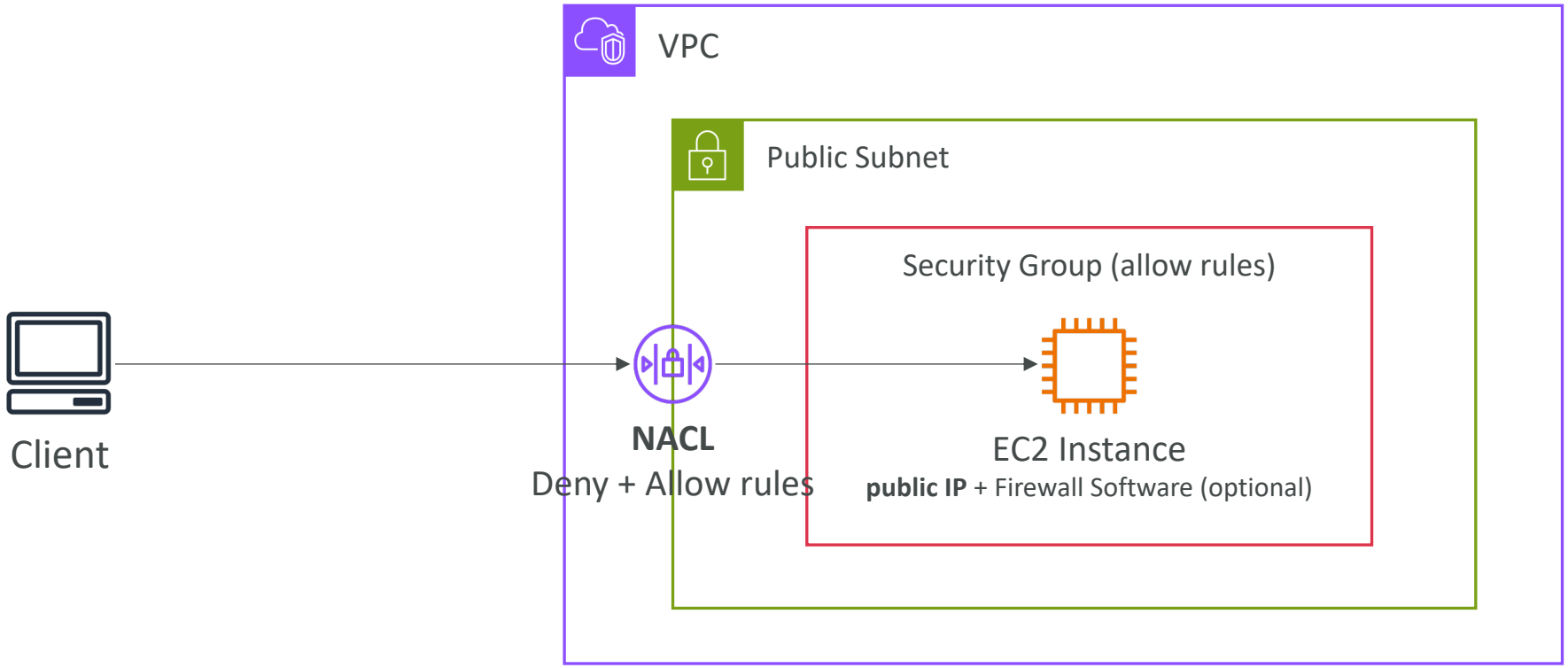
Caching Strategies



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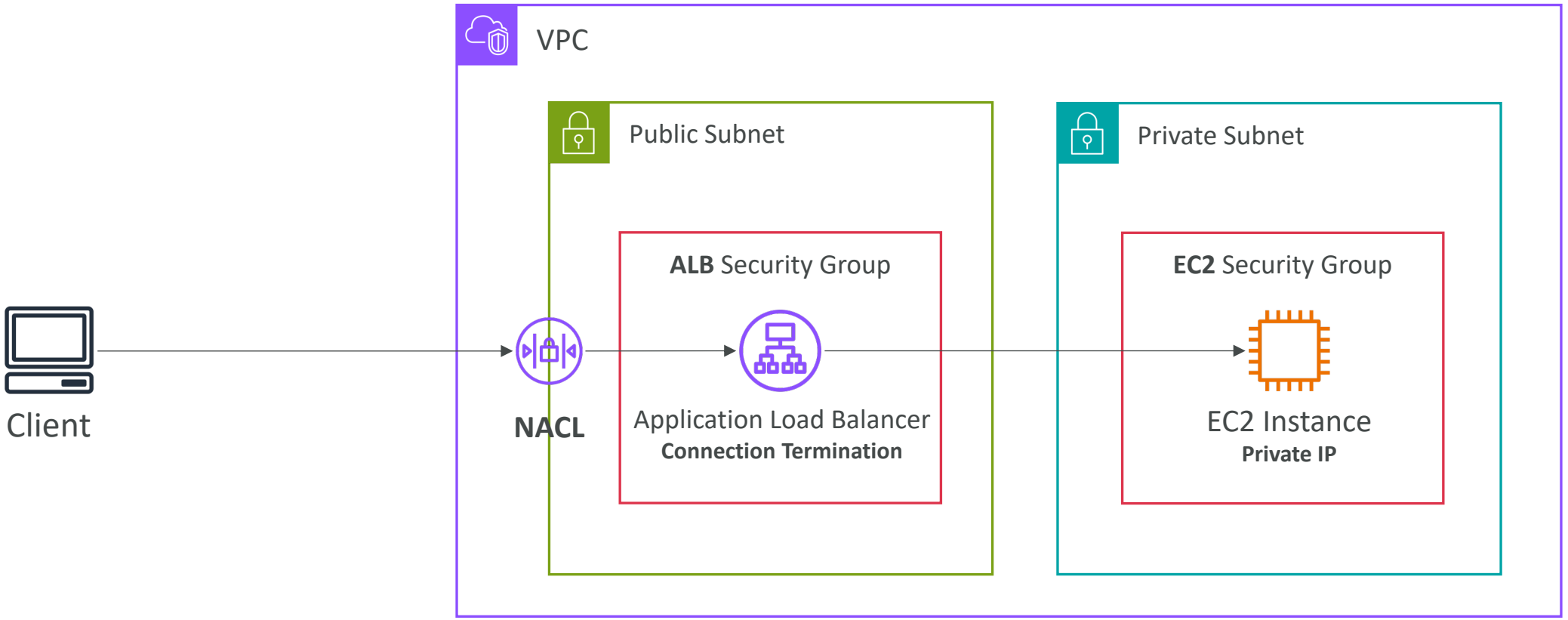
Blocking an IP address



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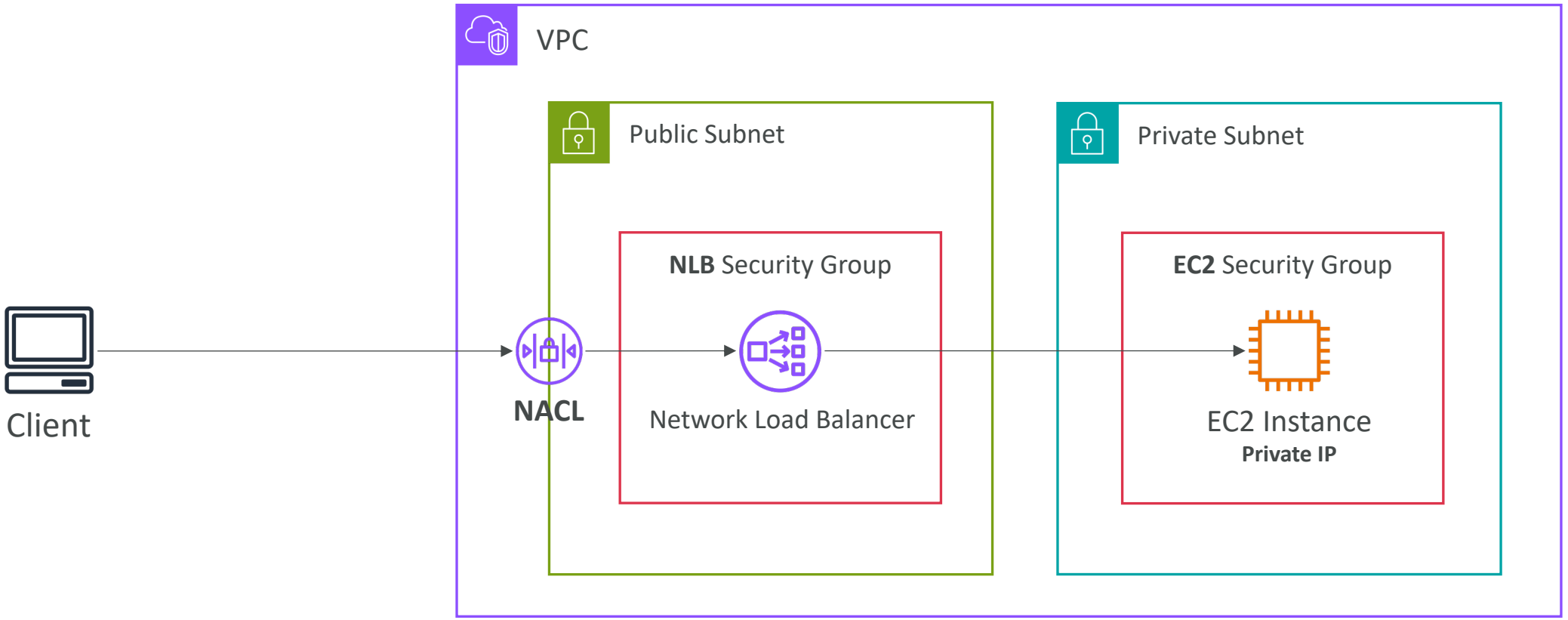
Blocking an IP address – with an ALB



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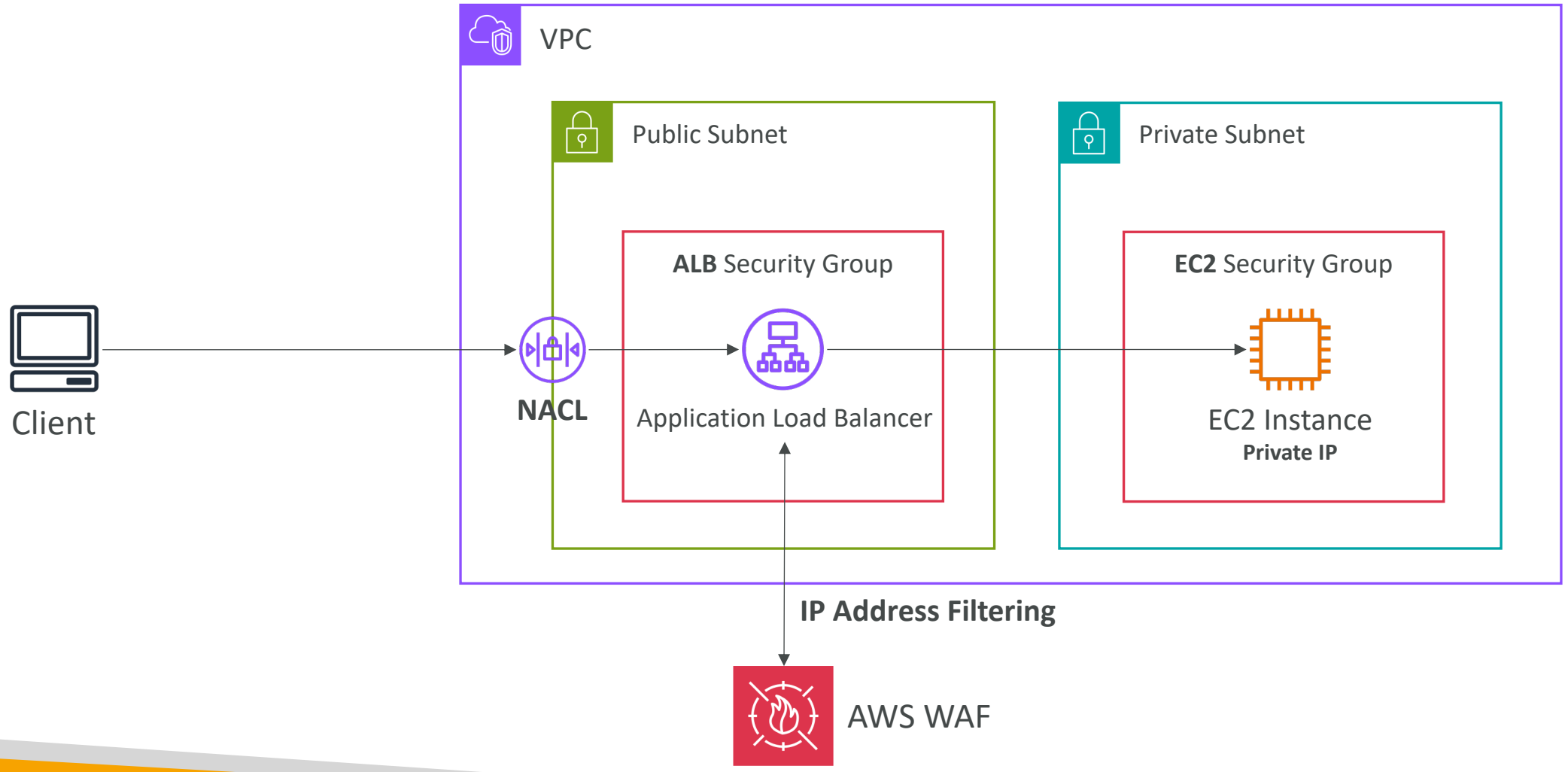
Blocking an IP address – with an NLB



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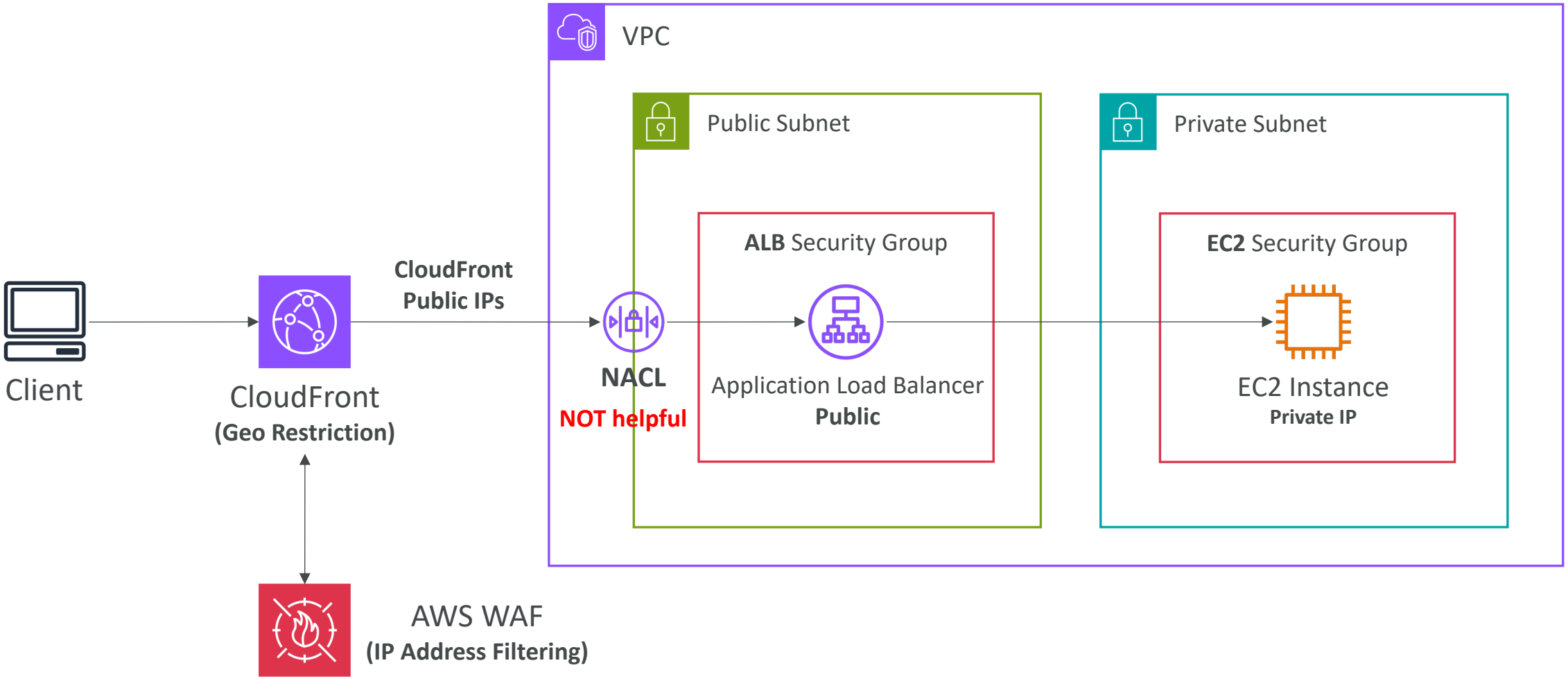
Blocking an IP address – ALB + WAF



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Blocking an IP address – ALB, CloudFront & WAF



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High Performance Computing (HPC)

- The cloud is the perfect place to perform HPC
- You can create a very high number of resources in no time
- You can speed up time to results by adding more resources
- You can pay only for the systems you have used

- Perform genomics, computational chemistry, financial risk modeling, weather prediction, machine learning, deep learning, autonomous driving

- Which services help perform HPC?

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Data Management & Transfer

- AWS Direct Connect:
  - Move GB/s of data to the cloud, over a private secure network
- Snowball & Snowmobile
  - Move PB of data to the cloud
- AWS DataSync
  - Move large amount of data between on-premises and S3, EFS, FSx for Windows

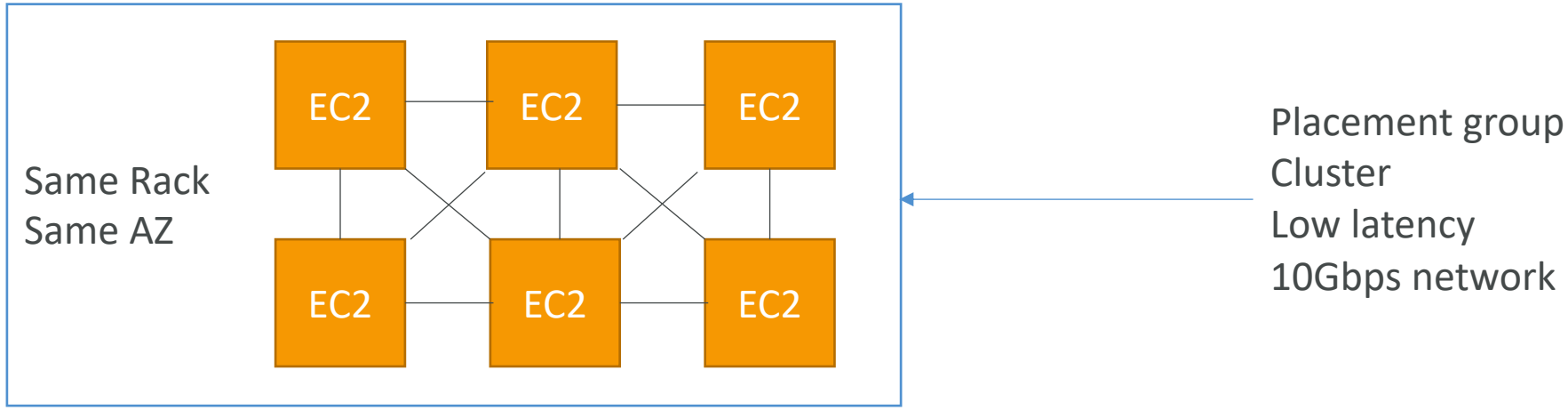
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Compute and Networking

- EC2 Instances:
  - CPU optimized, GPU optimized
  - Spot Instances / Spot Fleets for cost savings + Auto Scaling

- EC2 Placement Groups: **Cluster** for good network performance



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Compute and Networking

- EC2 Enhanced Networking (SR-IOV)
  - Higher bandwidth, higher PPS (packet per second), lower latency
  - Option 1: **Elastic Network Adapter (ENA)** up to 100 Gbps
  - Option 2: Intel 82599 VF up to 10 Gbps – LEGACY

- **Elastic Fabric Adapter (EFA)**
  - Improved ENA for **HPC**, only works for **Linux**
  - Great for inter-node communications, **tightly coupled workloads**
  - Leverages Message Passing Interface (MPI) standard
  - **Bypasses the underlying Linux OS to provide low-latency, reliable transport**

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Storage

- Instance-attached storage:
  - **EBS:** scale up to 256,000 IOPS with io2 Block Express
  - **Instance Store:** scale to millions of IOPS, linked to EC2 instance, low latency

- Network storage:
  - **Amazon S3:** large blob, not a file system
  - **Amazon EFS:** scale IOPS based on total size, or use provisioned IOPS
  - **Amazon FSx for Lustre:**
    - HPC optimized distributed file system, millions of IOPS
    - Backed by S3

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Automation and Orchestration

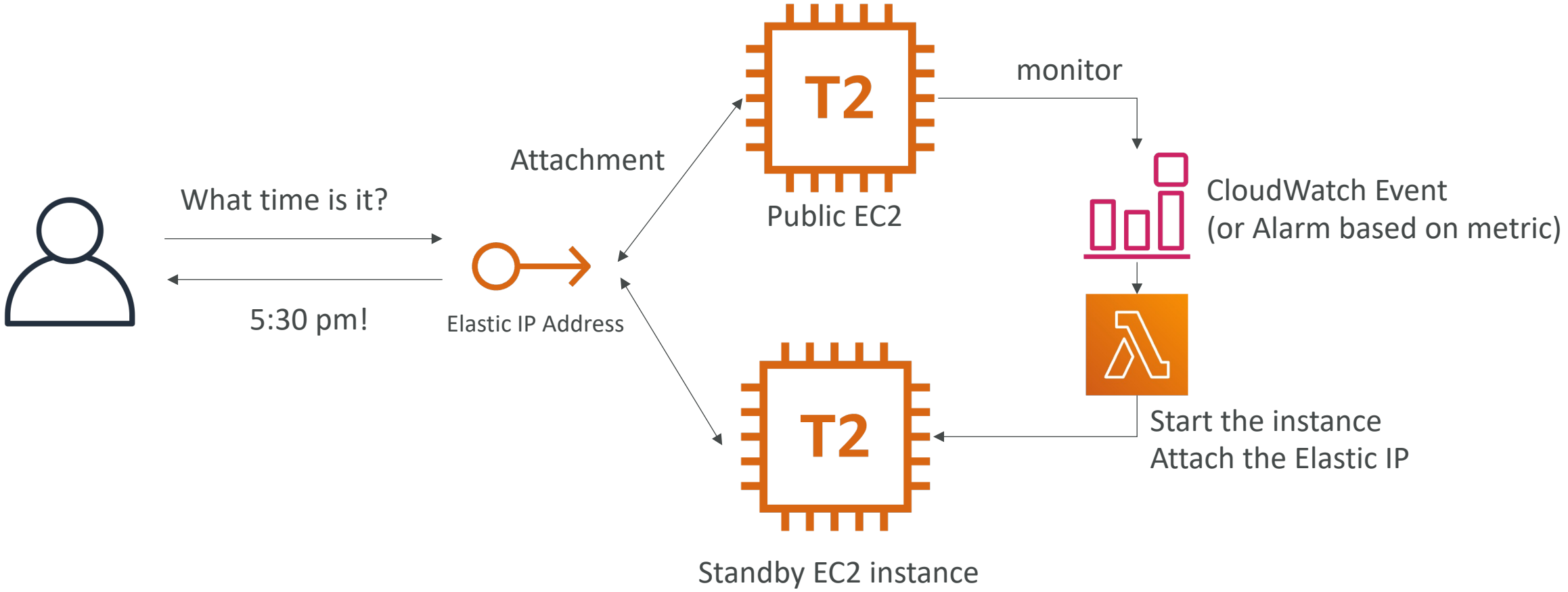
- AWS Batch
  - **AWS Batch** supports multi-node parallel jobs, which enables you to run single jobs that span multiple **EC2** instances.
  - Easily schedule jobs and launch EC2 instances accordingly

- AWS ParallelCluster
  - Open-source cluster management tool to deploy HPC on AWS
  - Configure with text files
  - Automate creation of VPC, Subnet, cluster type and instance types
  - **Ability to enable EFA on the cluster (improves network performance)**

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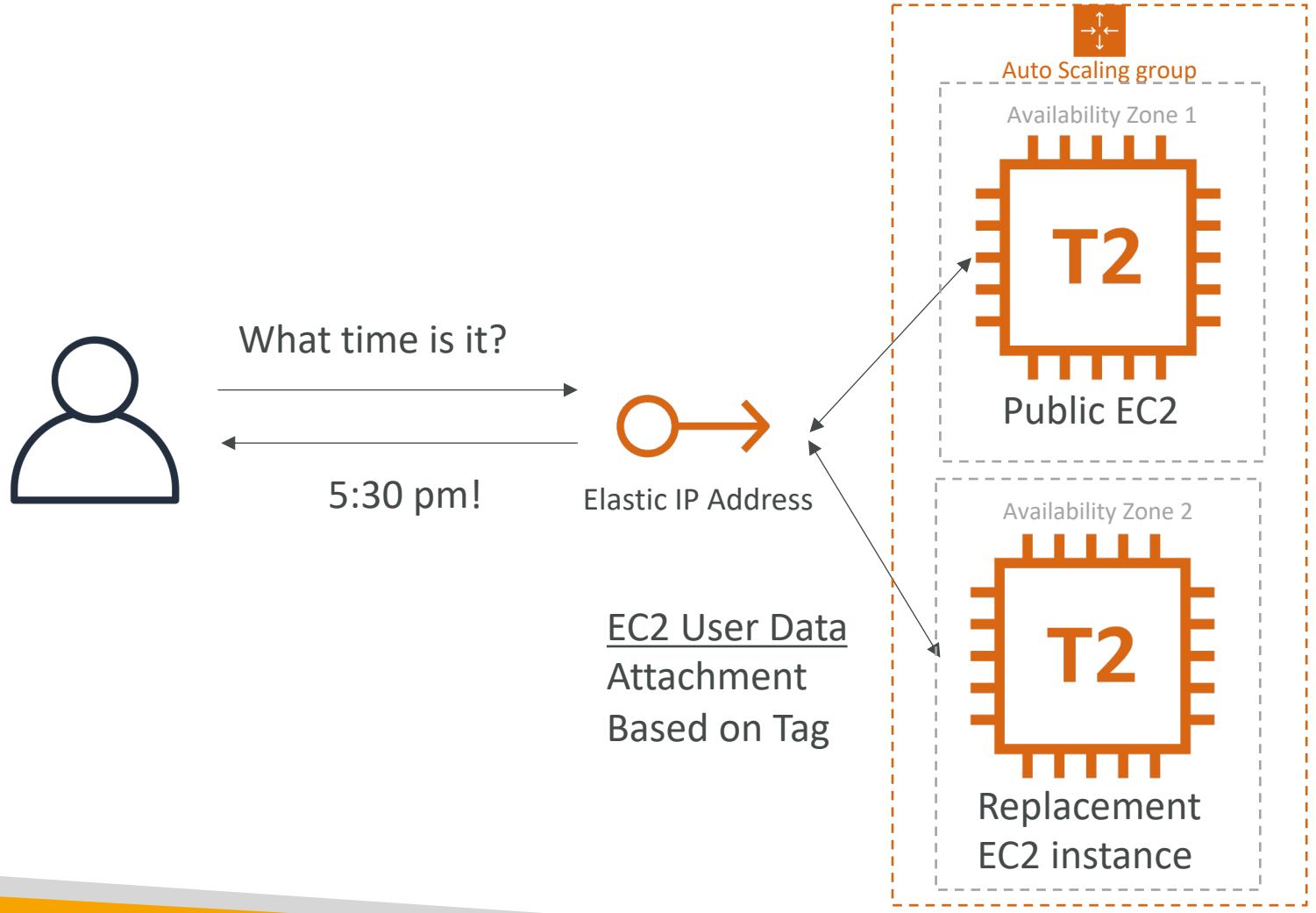
Creating a highly available EC2 instance



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Creating a highly available EC2 instance With an Auto Scaling Group



ASG Settings
1 min
1 max
1 desired
>= 2 AZ

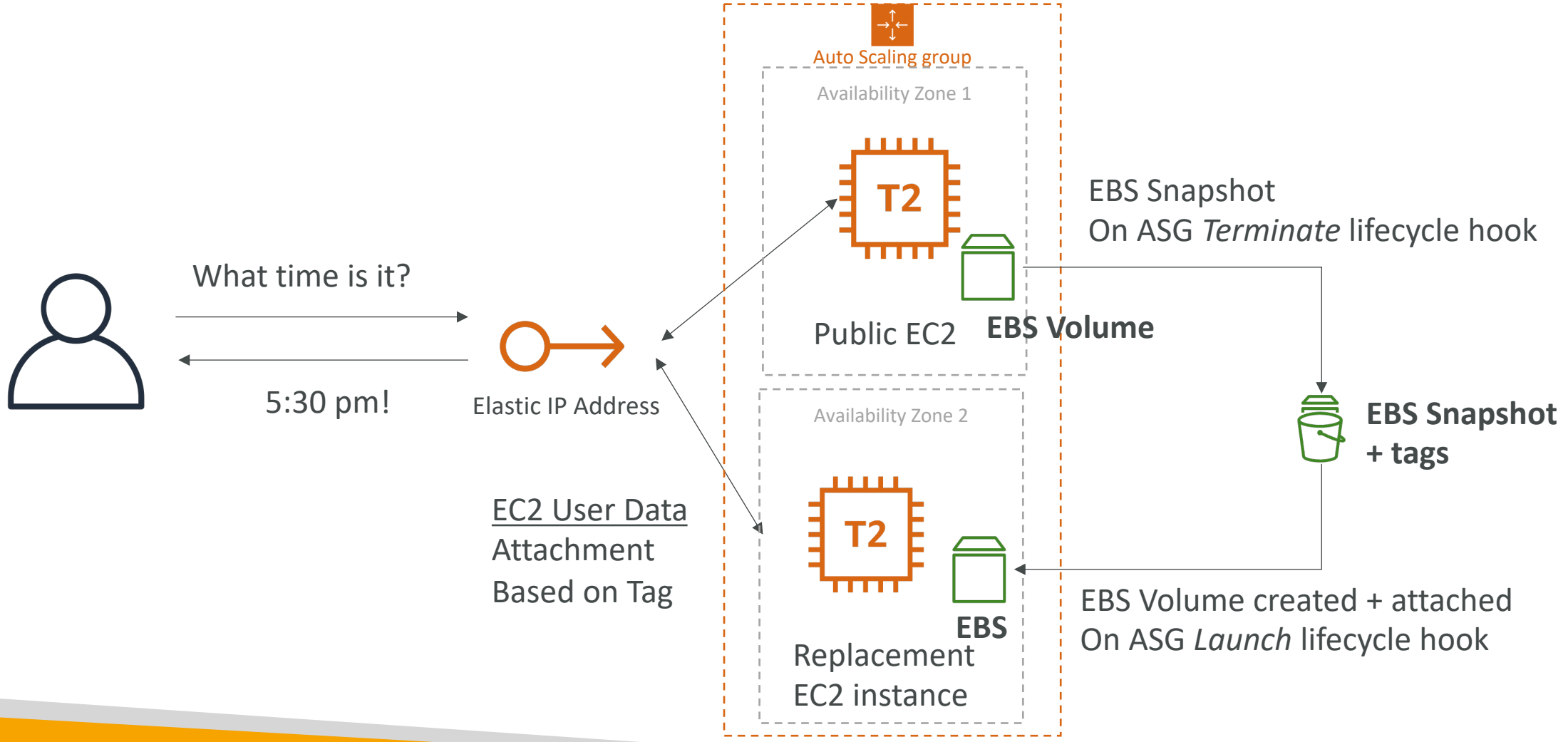
EC2 user data to attach
The Elastic IP

EC2 instance role to
Allow API calls to attach
The Elastic IP

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Creating a highly available EC2 instance With ASG + EBS



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